

Multiple Choice

1. **Answer:** D

Options: A) Alpha-ketoglutarate; B) Glutamate; C) Acetyl CoA; D) Oxaloacetate; E) Citrate

Note: Correct option: D - Oxaloacetate

2. **Answer:** A

Options: A) Apical tissue; B) Sclerenchyma tissue; C) Ground tissue; D) Vascular tissue; E) Cork tissue

Note: Correct option: A - Apical tissue

3. **Answer:** C

Options: A) carbon atoms; B) oxygen atoms; C) nitrogen atoms; D) hydrogen atoms

Note: Correct option: C - nitrogen atoms

4. **Answer:** A

Options: A) Animals use the sound of the ocean waves for directional cues in migration.; B) Animals rely solely on memory from previous migrations.; C) Animals migrate based on the patterns of the stars alone.; D) Animals follow changes in air pressure for migration.; E) Animals use temperature changes for migration.

Note: Correct option: A - Animals use the sound of the ocean waves for directional cues in migration.

5. **Answer:** C

Options: A) The suspensor is derived from the basal cell.; B) Cotyledons are derived from the apical cell.; C) Shoot apical meristem formation occurs after seed formation.; D) Precursors of all three plant tissue systems are formed during embryogenesis.

Note: Correct option: C - Shoot apical meristem formation occurs after seed formation.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'It must be present in a population before natural selection can act upon the population.'

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'It attracts predators away from the nestlings.'

Long Form Response

11. **Answer:** A shift in the balance of estrogen and progesterone occurs, the secretion of estrogen by the placenta rises sharply, oxytocin is released, relaxin loosens the connections between the bones of the pelvis, and prostaglandins stimulate the smooth muscle of the uterus.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Plants get rid of their waste products through diffusion via the pores of the leaves or root cells into the soil, and by shedding leaves.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key